

Laxmi Charitable Trust's
Sheth L.U.J College of Arts & Sir M.V. College of Science and Commerce
Department of Information Technology (Bsc.IT Semester III)
Python for Data Science Practical

Practical - II

Roll No. : S031	Name: Panvir
Class : SYIT	Batch : 01
Date of Assignment : 22-7-26	Date/Time of Submission : 22-7-26

Questions:

2a. Write a python code to create nested tuples.

Code:-

```
student=("panvir",19, ("bscit", "semIII"))
print("name: ",student[0])
print("Age: ",student[1])
print("Dept: ",student[2][0])
print("panvir S031")
```

Output:-

```
name:  panvir
Age:   19
Dept:  bscit
panvir S031
```

2b. Write a python code to sort the nested tuple using sorted () function.

Code:-

```
data = [(1,4), (3,2), (5,1)]

sorted_data = sorted(data, key=lambda x: x[1])

print(sorted_data)
|
print("panvir S031")
```

Output:-

```
=====
[(5, 1), (3, 2), (1, 4)]
panvir S031
|
```

2c. Write a python code to copy or clone list.

Code:-

```
list1=[1,2,3,4,5]
list2=list1
print(list2)|
print("panvir S031")
```

Output:-

```
----- r
[1, 2, 3, 4, 5]
panvir S031
|
```

2d. Write a python code to check immutability property of python tuples.

Code:-

```
name=("panvir","raju","krish")
name[-1]="shaam"
print("Panvir S031")|
```

Output:-

```
===== RESTART: D:/Panvir/Py for Datascience/prac2.py =====
Traceback (most recent call last):
  File "D:/Panvir/Py for Datascience/prac2.py", line 3, in <module>
    name[-1]="shaam"
TypeError: 'tuple' object does not support item assignment
```

Curiosity Questions:

Q1. Define a tuple. How is it different from a list? Give one example.

Q2. Write a Python program to create the following tuples:

- a) An empty tuple
- b) A tuple with integers only
- c) A tuple with mixed data types
- d) A nested tuple (tuple inside another tuple)

Q3. Write a program to access elements of a tuple using indexing and slicing.

Q4. Create a tuple and find the following:

- Length of the tuple
- Maximum and minimum value (if all are numbers)
- Count of a specific item

Q5. Write a program to convert a list into a tuple.

Q6. Write a program to sort a tuple of numbers in ascending order.

Q7. Write a program to show that tuples are immutable (cannot be changed).

Q8. Write a program to clone (copy) a list and print both original and copied lists.

Q9. Write the output of the following code:

```
t = (1, 2, 3, 4, 5)
```

```
print(t[1:4])
```

Q10. Create a tuple of student marks: (78, 85, 92, 88, 76). Write a program to:

- Find average marks
- Display marks above 80